Teamwork Specification: Sprint 1

Courier Management System (CMS) — Frontend & System Requirements

1. Introduction

The primary objective of this teamwork project is to develop a Web-based Enterprise Information System (IS) for courier and logistics management. Modern delivery and fleet operations demand centralized digital platforms to streamline shipment bookings, optimize driver routing, and ensure real-time supply chain transparency. The project is structured across two major iterations (Sprint 1 and Sprint 2). In this first phase (Sprint 1), teams must define the global system requirements, create the conceptual database architecture (Entity-Relationship ER Diagram), and implement a functional, **navigable static user interface prototype** utilizing standard Web technologies (HTML5, CSS3, and JavaScript form validations).

2. Description

The Courier Management System (CMS) manages the end-to-end lifecycle of parcel deliveries, fleet assets, and logistics personnel. The system coordinates interactions across four core user roles: Clients, Couriers, Staff and Administrators (for now on, refer as Admin). It enforces physical load constraints and regulatory compliance across all operations.

2.1. Global Needs

- Clients: Users who request delivery services. They register products and place orders specifying Origin and Destination locations using individual GPS coordinates (Latitude and Longitude). They can access the status of an active order or access completed orders (historic data).
- Parcel deliveries (or orders): An order has an origin and Destination, one or more products involved, a courier that makes the order, a vehicle that can accommodate its products and an order status.
- Couriers: Field drivers managing one or more vehicles. Regarding routes/orders, a courier can:
    1) Select appropriate vehicle for that order;
    2) selects and executes one route/order at a time ('single active task' constraint) or can consolidate multiple active orders into a single route;
    3) View completed orders.

  They transition order status sequentially (Initiated -> In Progress -> Completed/Cancelled). Upon completion, the courier must submit the mandatory Electronic Proof of Delivery (e-POD),

consisting of a **digital Canvas signature** or **a delivery photo**. Upon cancellation, the order returns to the list of orders.

- Staff: Backoffice administrators responsible for system governance. Staff members validate courier accounts and approve individual vehicles before allocation (verifying valid insurance and inspection certificates). Staff receive an alert when the information is updated/outdated (i.e. when a new inspection certificate is uploaded or when it is outdated).
- Admin: Administrators are responsible for validating new staff members, having access also to a dashboard. This dashboard is responsible for showing statistic information, such as, number of couriers, staff, clients, routes statis and other operational analytics dashboards.
- Other information that is relevant to the operation of the proposed IS. **The addition of features that prove important to the IS will be valued**.

2.2. OpenStreetMap (Leaflet.js API) Requirement

To embed geospatial intelligence into the platform, the frontend must integrate interactive mapping powered by the open-source Leaflet.js library and OpenStreetMap (OSM) tiles:

1. Client Order Map: The order creation form must feature an embedded interactive map container. For this Sprint 1, Leaflet.js renders one marker in a random place, for example, on FEUP.
2. Courier Navigation Cockpit: When a courier views an active order, the system displays the OpenStreetMap with the previous marker (for example, FEUP marker).

3. Minimal System (Required Static Pages)

The static frontend prototype must consist of the (minimum) following interconnected HTML5 documents:

HTML Page	Target User Role	Key Features & Functional Requirements
Home / Portal	Public	Overview some metrics, service highlights, unified access to login and registration.
Login/Register Area	Public	A specific area for users to register or make the login.
Client Page	Client	Order & product submission forms with GPS input fields and interactive OpenStreetMap (Leaflet.js) container; Order status for accepted orders.

HTML Page	Target User Role	Key Features & Functional Requirements
Courier Page	Courier	Active vehicle selector, map visualization (in this sprint, only one marker placement is needed, for example, a marker on FEUP); sequential status transition buttons (Initiated -> In Progress -> Completed/Cancelled); Proof of delivery signature or photo upload simulator. List of completed orders.
Staff Panel	Staff	Pending courier & vehicle approvals (with visual alerts for expired insurance/inspection).
Admin Panel	admin	Pending staff approvals; edit personal information and access Relevant operational KPI statistics dashboard.
Team Page	Public	Academic identification of all group members (Name, student ID, institutional email, course, team number).

4. Considerations about Information (Conceptual ER Model)

Teams must design a normalized Entity-Relationship (ER) conceptual model to support database persistence. The conceptual schema must cover (**but not limited to**) the following core domain entities:

Entity	Primary Attributes	Domain Description & Role
User	username, password_hash, email, name, phone, nif, role, created_at	Unified user entity supporting Staff, Courier, admin and Client roles.

Entity	Primary Attributes	Domain Description & Role
Vehicle	license_plate, model, max_weight_kg, max_vol_m3, insurance_expiry_date, inspection_expiry_date, insurance_expiry_doc, inspection_expiry_doc, created_at	Fleet assets registered by Couriers and audited by Staff.
Vehicle_Type	type_name	Standard vehicle categories (Bike, Moto, Van, Truck) for capacity checks.
Product	description, category, weight_kg, volume_m3, created_at	Cargo items with physical weight and volume dimensions.
Order	origin_lat, origin_long, dest_lat, dest_long, status, attempt, created_at	Delivery service request with GPS route coordinates.
Route	status	Delivery assignment linking courier, vehicle, and order.

5. Development Process and Technologies

Teams are made up of **4 members** (5 elements should be possible also, depending on the arrangement of teams in each course). The development process must be iterative.

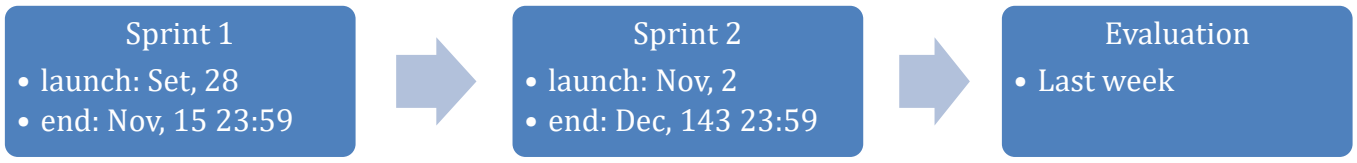


Figure 1: Development process and dates.

Process for Sprint 1:

1. In the initial phase, **definition of requirements, use cases and layout of pages** by the entire team;
2. It is mandatory (and beneficial) to **identify a team leader** to keep the project on track by delegate tasks according to individual strengths and ensuring deadlines are met. They also act as the central point of communication, resolving conflicts early to maintain a cohesive and high-quality final submission.
3. Then, **divide the features/pages by the team** (Person A will be responsible for page X; Person B will be responsible for page Y, etc);
4. **Creation of the conceptual model** (ER model) to support the project.

Development must follow an agile, iterative methodology. Use of GIT version control and GitHub is mandatory. The repository name must follow the format: EIS_2627-GXXX_MXXX (e.g. EIS_2627-G01_MEEC). All team members must contribute meaningful commits across both sprints. Form validation via JavaScript is mandatory: empty required fields must prevent form submission and display popup/text error alerts. Netbeans should preferably be used as the IDE, but other can be used (VSCode, etc).

Additional guidance, or new requirements, may be given during the project.

6. Evaluation & Submission

Sprint 1 accounts for 30% (Project) + 2.5% (Academic Report). Submission Deadline **on Moodle**: **November 15 at 23:59 (Moodle Time)**.

- Deliverables: A single .zip file containing:
    1) HTML/CSS/JS source code with sample test data
    2) High-resolution Entity-Relationship (ER) Diagram image
    3) Self and hetero evaluation using the official Moodle template, and
    4) Academic Report using the official Moodle template. This report must be in .pdf.
- Only **the team leader** of each team **submits the .zip file on Moodle system**.

Project Evaluation Component	Weight	Key Quality Criteria
ER Conceptual Model	30%	Correct modeling of entities and relationships between them.

Project Evaluation Component	Weight	Key Quality Criteria
Frontend Prototype (HTML/CSS)	45%	Responsive design, professional visual hierarchy, fluid inter-page navigation.
Interactive Map (Leaflet/OSM)	15%	Correct Leaflet.js mapping, marker placement from GPS coordinates, route polyline.
JavaScript Validations	10%	Client-side empty field prevention, numeric coordinate checks, user feedback.

7. Version Control

Version	Release Date	Main Specification Changes
V1.0	Sept 28, 2026	Initial specification release.